participants to game downgrades in order to "pick up spreads," creating additional inefficiencies. Many fixed-income indexes rely on ratings as an inclusion factor, and several types of institutional investors are required, by mandate or regulation, to restrict themselves to bonds over a certain ratings threshold. Putting all these factors together, there is forced selling on downgrades by passive vehicles on the date of an index exclusion. This puts pressure on the price of the assets even when the information driving the downgrade has been fully processed. These well-documented inefficiencies lead to another set of dislocations where institutional investors try to predict behavior that is noneconomic, and the participant is worse off on a net present value basis.

Another frequent culprit is the presence of noneconomic or constrained economic agents in a market. Often this takes the form of some sort of official sector intervention program with strictly defined criteria for inclusion. The ECB has employed many such programs, including the previously mentioned corporate sector buyback program and others targeting an array of asset types. The assets that are included frequently become displaced relative to otherwise similar assets outside the scope of such a program. Other times it is a matter of habitat preference for certain investors. There are investors who either must for regulatory reasons, or merely prefer to for ease of operations, own bonds that are denominated in their own currency, traded in their home market, or issued in their own legal jurisdiction. Such investors will care little if bonds of the same issuer in other currencies trade significantly rich or cheap, relative to the bonds they own. In times of market stress, many of these factors act in concert to amplify these dislocations.

In the modern financial world, all of these kinds of forces are being exerted across many time zones and regulatory regimes, and there are obvious short circuits that can sever the market prices of instruments reflecting the same underlying risk. One might be able to observe the risk of a given company through its debt, its CDS, the cash and